

Predictive Analytics Poster Rubric (Poster-Only Assessment)

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
1	Prediction Goal(s)	5	5 pts: Clear, specific predictive objective(s); response type (regression/classification) explicitly identified and justified; strong business/research rationale.	3–4 pts: Goal stated; response type identified but weakly justified; limited link to context.	0–2 pts: Goal unclear/missing; response type unspecified; no rationale.
2	Dataset Overview	8	7–8 pts: Reports #obs/#features; types for key features; response variable identified/typed; data provenance/window stated.	4–6 pts: Size and response identified; partial feature typing; minimal provenance.	0–3 pts: Missing size/response ID; little/no feature typing or context.
3	Exploratory Data Analysis (EDA)	12	10–12 pts: Appropriate summaries (response + key predictors); ≥ 2 relevant plots; interprets skew/outliers/imbalance/seasonality; connects insights to modeling choices and metric selection.	6–9 pts: Basic stats and ≥ 1 plot; some interpretation but limited linkage to modeling.	0–5 pts: Incomplete stats; weak/irrelevant visuals; minimal/no interpretation.
4	Data Preparation (Feature Engineering, Missing Data, etc.)	20	17–20 pts: ≥ 1 non-trivial new feature (interactions, lags/rolling stats, domain-driven ratios, text-derived signals) with clear rationale; quantifies missingness by feature; justified strategy (drop/impute/advanced) with leakage-safe implementation; demonstrates impact via ablation/CV deltas or sensitivity to imputation choices.	10–16 pts: New feature with partial rationale; identifies missingness and applies a reasonable method; limited evidence of impact or sensitivity analysis.	0–9 pts: No new/trivial feature; missingness unreported or mishandled; unjustified/inappropriate method; no evidence of value.
5	Baseline Model (Choice, CV, Interpretation)	20	17–20 pts: Appropriate simple baseline (e.g., Linear/Logistic Regression); pipeline is leakage-safe; $k=5/10$ CV with correct metric (RMSE/MAE/Accuracy/F1/AUC); tracks mean/variance; summarizes implications and limitations; proposes concrete next steps.	10–16 pts: Reasonable model; CV present with partial integration/details; metric adequate; interpretation limited or generic.	0–9 pts: Inappropriate/unspecified baseline; no/incorrect CV; unclear metric; raw numbers only; no reflection.
6	More Complex Model (Choice, Tuning, Comparison)	25	22–25 pts: Model aligned to data/goal (RF/GBM/XGBoost/SVM/regularized GLM/NN); hyperparameter search (grid/random/Bayesian) inside proper CV; leakage controls (scaling/encoding inside folds); comparative results (tables/plots) with variability and selection criterion; rigorous comparison to baseline with practical interpretation (effect sizes, trade-offs), plus error/confusion analysis where relevant.	13–21 pts: Plausible model; some tuning with partial documentation; CV present but thin; basic comparison to baseline; limited practical takeaways.	0–12 pts: Misaligned/unspecified model; superficial/absent tuning; incorrect CV; opaque selection; no clear comparison or interpretation.
7	Report Quality & Visual Communication	10	9–10 pts: Clear, scannable poster; logical flow (Intro→Methods→Results→Conclusions); legible, labeled figures/tables (titles, axes, units, legends); consistent typography/layout; code refs/appendix where appropriate; minimal grammar issues.	6–8 pts: Generally readable; minor layout/labeling gaps; some figures under-explained or styling inconsistencies.	0–5 pts: Disorganized; illegible/unlabeled visuals; weak linkage between text and graphics; notable grammar/formatting issues.